Signal-Conditioned Memory Compression for Multi-Turn Conversations

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Abstract

Multi-turn conversation compression must balance two distinct objectives: preserving the geometric structure of the model's hidden-state representation (geometric fidelity) and maintaining behavioral output quality (answer likelihood). We show these objectives can diverge substantially: LongLLMLingua, a compression method optimized for answer likelihood, incurs logit ℓ_2 penalties of +103–161 units relative to uniform sampling across three budget levels while providing only marginal negative log-likelihood (NLL) improvement on 1,000 Multi-Session Conversations (163,755 evaluation rows). We introduce the Keep/Compress/Drop (kcd) codec, a sparse ternary turn-selection framework, and study three scoring signals: geometry-based (geo-kcd), semantic-based (sem-kcd), and query-conditioned geometry inside a semantic shortlist (sem-qcg). On Multi-Session Conversations (1,000 conversations, Qwen2.5-1.5B-Instruct), sem-qcg achieves logit ℓ_2 of 705.5 at budget 0.50—lower than uniform sampling (761.6)—while simultaneously improving answer NLL versus both semantic and geometry variants ($\Delta{-}109.4$ logit ℓ_2 and $\Delta{-}0.044$ NLL versus sem-kcd, both $p<0.001$ at conversation level). On a synthetic constraint-critical stress set, geometry-based scoring outperforms semantic scoring by up to +1.1 nats NLL ($p<0.001$, 3B scale), an effect that grows with model size. We characterize two scope conditions: the geometry advantage is architecture-dependent (present on Qwen2.5-{1.5B,3B} but reversed on Llama-3.2-3B) and length-dependent (degrades on 400–600 turn conversations). These findings establish geometry-behavior divergence as a consequential phenomenon and demonstrate that the choice of turn scoring signal is a first-class design decision in conversational memory compression.

1 Introduction

Deployed conversational systems face a hard constraint: as sessions grow, the full conversation history quickly exceeds a fixed context window. This is not merely a capacity problem: even when context fits, language models struggle to uniformly exploit information distributed across long histories, with performance degrading sharply for content appearing in the middle (Liu et al., 2024). Memory compression—selecting, condensing, or dropping prior turns to stay within a token budget—is therefore a core engineering challenge. An alternative is retrieval augmentation (Lewis et al., 2020) or hierarchical memory management (Packer et al., 2023), but these require external indexes or architectural changes; we focus on in-context compression that operates purely at inference time. Most prior compression work treats this as a single-objective problem: minimize the loss on held-out continuations, or equivalently, preserve behavioral quality as measured by answer NLL (Jiang et al., 2024; Pan et al., 2024).

*Code, data, and experiment logs: https://github.com/SteveMama/rt-geometry-memory.

We argue this framing is incomplete. Two distinct quantities are affected by compression: the behavioral quality of model outputs (answer likelihood, task accuracy), and the geometric fidelity of the model's hidden-state trajectory, which reflects how faithfully the compressed context reconstructs the internal representational state that the full context would produce. These objectives are not equivalent and, as we show, can diverge dramatically.

Geometry-behavior divergence. Our first observation is that LongLLMLingua—a strong baseline optimized for NLL—achieves only marginal behavioral improvements (-0.07 to -0.18 nats versus uniform sampling) while incurring catastrophic geometric degradation ($+103$ to $+161$ logit ℓ_2 units on MSC, n=1,000 conversations). A method that looks adequate by behavioral metrics may be distorting the model's representational geometry substantially.

The kcd codec. We introduce a sparse ternary codec that assigns one of three actions to each prior turn—keep, compress, or drop—under a fixed token budget. The codec is signal-agnostic: any per-turn score can rank turns for retention. We study three signals: (i) geometry-based: transported hidden-state curvature and subspace energy, (ii) semantic: cosine similarity of turn hidden states to the target turn, and (iii) query-conditioned geometry (sem-qcg): geometric features projected through the hidden state at the current query turn, applied as a reranker within a semantic shortlist.

Main findings. On Multi-Session Conversations (Xu et al., 2022), sem-qcg is the only variant achieving simultaneously best geometric fidelity and best behavioral quality at budget ≥ 0.35, resolving the geometry-behavior tradeoff that plagues simpler signals. At budget 0.50, sem-qcg achieves lower logit ℓ_2 than uniform random sampling—meaning the policy is more faithful to the full-context hidden state than retaining an arbitrary subset of turns. On a constraint-critical stress set (exact user instructions, retrieval packets, code constraints), geometry-based scoring beats semantic scoring on both metrics, with the NLL damage from using the wrong signal growing from $+0.63$ nats at 1.5B to $+1.13$ nats at 3B.

Contributions.

1. We demonstrate geometry-behavior divergence at scale: methods optimizing NLL can catastrophically degrade hidden-state geometric fidelity, motivating dual-objective evaluation of memory compression.
2. We introduce sem-qcg and show it is the only kcd variant achieving simultaneously best geometric fidelity and behavioral quality on msc (both p<0.001, conversation-level signflip test).
3. We provide a support-turn rescue mechanism account explaining why geometry works on constraint-critical memory: user instructions produce larger hidden-state displacement than semantically similar assistant echoes (17/36 conversations rescued by geometry versus 7/36 by semantic).
4. We characterize two scope conditions (architecture-dependence and conversation length-dependence) that bound where geometry-based signals add value.

The rest of the paper is organized as follows. Section 2 introduces the geometric framework and kcd codec. Sections 3–6 present experiments. Section 7 covers related work. Section 9 states limitations explicitly.

2 Background

2.1 Conversation State Geometry

Let $\mathbf{h}_t \in \mathbb{R}^d$ denote the last-layer hidden state produced by a causal language model after processing the t-th conversation turn, normalized to the unit hypersphere $\mathbb{S}^{d-1}$. The sequence $(\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_T)$ traces a trajectory on $\mathbb{S}^{d-1}$.

Compact structure. We quantify intrinsic dimensionality via Rank-95: the number of principal components explaining 95% of variance in the centered trajectory matrix, normalized by the sequence length (Roy and Vetterli, 2007). We observe that real conversation trajectories are highly compact (Rank-95 $\approx$ 1.05–1.53), whereas shuffled controls—which break temporal ordering—achieve Rank-95 $\approx$ 1.46–1.80 (Table 10). Low intrinsic dimensionality is a well-established property of transformer representations (Li et al., 2018; Aghajanyan et al., 2021) and of contextual embeddings more broadly (Ethayarajh, 2019; Cai et al., 2021); our finding extends this to the temporal trajectory of conversation-state hidden states, which is both more compact and more structured than a shuffled baseline. This motivates transported geometry as a discriminative signal.

Geometric distortion predicts decoder drift. For a compressed context c_π under compression policy π, define the logit ℓ_2 between the full and compressed contexts as:

$$\mathrm{LL2}(\pi) = \|\mathbf{e}_{\text{full}} - \mathbf{e}_\pi\|_2$$

where $\mathbf{e}$ denotes the output logit vector at the prediction position. We find that the Pearson correlation between per-turn geometric distortion and logit ℓ_2 is $\rho = 0.989$–0.994 across all tested models ($p < 0.001$ in all cases; Table 10). This tight correspondence motivates geometric distortion as a proxy for representational harm.

Transported geometry. We use parallel transport on $\mathbb{S}^{d-1}$ to bring per-turn displacement vectors into a common tangent space. For consecutive hidden states $\mathbf{h}_t, \mathbf{h}_{t+1}$, the incremental displacement in the tangent space at $\mathbf{h}_t$ is:

$$\mathbf{u}_t = \mathrm{PT}_{\mathbf{h}_t \to \mathbf{h}_{\text{ref}}}\big(\mathrm{Log}_{\mathbf{h}_t}(\mathbf{h}_{t+1})\big)$$

where $\mathrm{Log}_{\mathbf{h}_t}(\cdot)$ is the Riemannian logarithm map and $\mathrm{PT}_{\mathbf{h}_t \to \mathbf{h}_{\text{ref}}}(\cdot)$ is parallel transport to a reference point $\mathbf{h}_{\text{ref}}$ taken as the Fréchet mean of the trajectory (Afsari, 2011). Per-turn geometric features—curvature, subspace energy, alignment—are derived from $\mathbf{u}_t$.

2.2 The kcd Codec

Given a conversation with T turns and a token budget B, the kcd codec assigns each turn $t \in \{1, \ldots, T\}$ one of three actions:

$$a_t \in \{\text{keep}, \text{compress}, \text{drop}\}$$

subject to the constraint:

$$\sum_{t=1}^{T} \text{cost}(a_t, \tau_t) \leq B$$

where τ_t is the token count of turn t and $\text{cost}(\cdot)$ satisfies $\text{cost}(\text{keep}, \tau) = \tau$, $\text{cost}(\text{compress}, \tau) = \lceil \alpha\tau \rceil$ for compression ratio $\alpha < 1$, and $\text{cost}(\text{drop}, \tau) = 0$.

The codec is parameterized by a per-turn score s_t and applies a greedy assignment: turns with highest scores are kept first, followed by compressed turns, and remaining turns are dropped. The budget B is expressed as a fraction of the full context length (we evaluate at $B \in \{0.20, 0.35, 0.50\}$).

2.3 Scoring Signals

We study three choices of s_t:

geo-kcd. The score is a linear combination of transported geometric features: curvature magnitude $\kappa_t = \|\mathbf{u}_t\|$, subspace energy σ_t (proportion of variance captured by $\mathbf{u}_t$), and alignment $\alpha_t = \cos(\mathbf{u}_t, \bar{\mathbf{u}})$ with the mean displacement.

$$s_t^{\text{geo}} = w_1\kappa_t + w_2\sigma_t + w_3\alpha_t$$

Weights are set by a held-out validation search on MSC. Geometric risk is higher at turns where the trajectory deviates strongly, which empirically corresponds to turns carrying constraint-critical information (user instructions, code snippets, factual assertions).

sem-kcd. The score is the cosine similarity of the turn's hidden state to the target turn (the turn whose continuation we are predicting):

$$s_t^{\text{sem}} = \cos(\mathbf{h}_t, \mathbf{h}_{\text{target}})$$

This favors turns whose surface content is semantically similar to the current query, as measured in the model's representation space.

sem-qcg (Query-Conditioned Geometry). This two-stage signal first builds a semantic shortlist of the top-k turns by s_t^{sem}, then reranks within that shortlist using geometry conditioned on the hidden state at the query turn:

$$\mathbf{u}_t^{(q)} = \mathrm{PT}_{\mathbf{h}_t \to \mathbf{h}_q}\big(\mathrm{Log}_{\mathbf{h}_t}(\mathbf{h}_q)\big)$$

where $\mathbf{h}_q$ is the hidden state at the query (target) turn. The reranking score is the projection of $\mathbf{u}_t^{(q)}$ onto the query-turn direction, capturing how geometrically proximate turn t is to the current query's representational context. Intuitively, sem-qcg biases retention toward turns whose information is load-bearing for the current query's hidden-state trajectory.

3 Geometry-Behavior Divergence

3.1 Setup

We evaluate on the Multi-Session Conversations (MSC) valid split (Xu et al., 2022), using 1,000 conversations and the Qwen2.5-1.5B-Instruct model (Qwen Team, 2025). This produces 163,755 evaluation rows. Baselines include uniform random turn sampling (uniform), recency-based retention (recency), recency scoring inside the kcd codec (recency-kcd), lexical TF-IDF (TF-IDF), and LongLLMLingua (Jiang et al., 2024).

We report two metrics: Logit ℓ_2 (lower is better geometric fidelity) and Answer NLL (lower is better behavioral quality), computed as the negative log-likelihood of the ground-truth assistant response tokens under the compressed context.

Table 1: External baseline comparison on MSC valid (n=1,000 conversations, 163,755 evaluation rows, Qwen2.5-1.5B). Bold = best per column. Lower is better for both metrics.

	Logit $\ell_2 \downarrow$			Answer NLL $\downarrow$		
Policy	B=0.20	B=0.35	B=0.50	B=0.20	B=0.35	B=0.50
Uniform	627.2	613.8	596.3	3.361	3.269	3.171
Recency	605.3	572.6	537.5	3.236	3.050	2.901
Recency-kcd	604.9	590.9	564.4	3.219	3.021	2.860
TF-IDF	617.4	610.5	594.0	3.249	3.074	2.930
LongLLMLingua	730.4	741.0	757.2	3.114	3.070	3.013

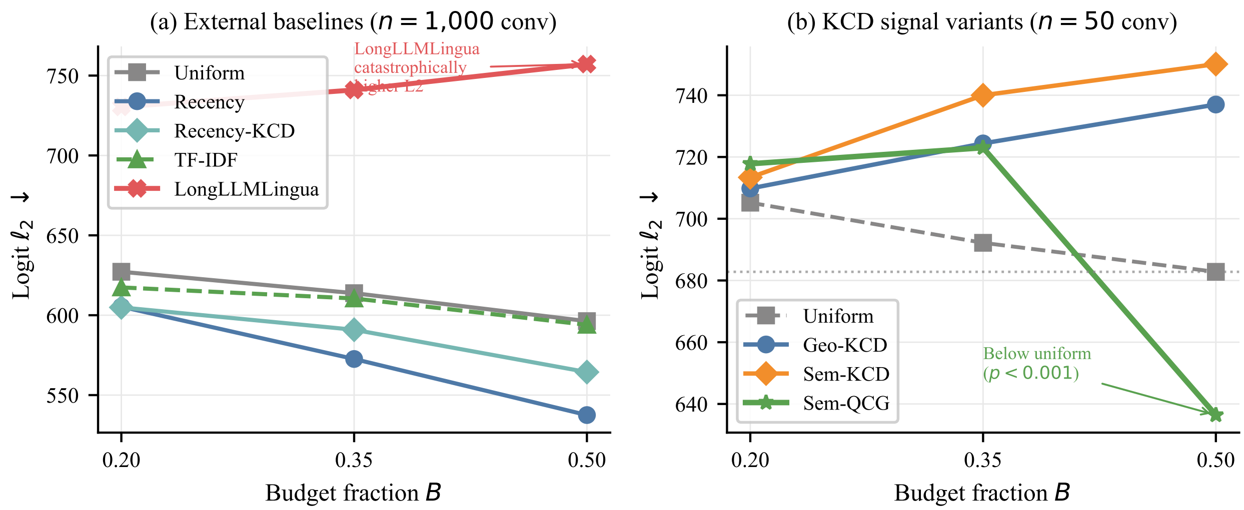

Figure 1: Logit ℓ_2 vs. budget for all policies. Left: external baselines on MSC (n=1,000 conv). LongLLMLingua incurs +103–161 logit ℓ_2 above uniform despite modest NLL gains. Right: KCD signal variants (n=1,000 conv). Sem-QCG achieves 705.5 at B=0.50—below the uniform baseline of 761.6 (dotted line).

3.2 Results

Table 1 shows a striking dissociation. LongLLMLingua achieves the lowest NLL at budget 0.20 (3.114 vs. uniform 3.361) but the highest logit ℓ_2 across all budgets (730.4–757.2 vs. uniform 596.3–627.2). The geometric degradation is +103 to +161 units above uniform—a method designed to preserve behavioral quality systematically damages the model's representational geometry.

Recency-kcd achieves the best NLL (3.219 at B=0.20), while plain recency achieves the best logit ℓ_2 (537.5 at B=0.50). No single policy dominates both objectives.

Implication. Evaluating compression methods on NLL alone is insufficient. A method can be behaviorally adequate while substantially distorting the hidden-state representation the model uses for reasoning, generation planning, and chain-of-thought. This motivates dual-objective evaluation, which we adopt throughout.

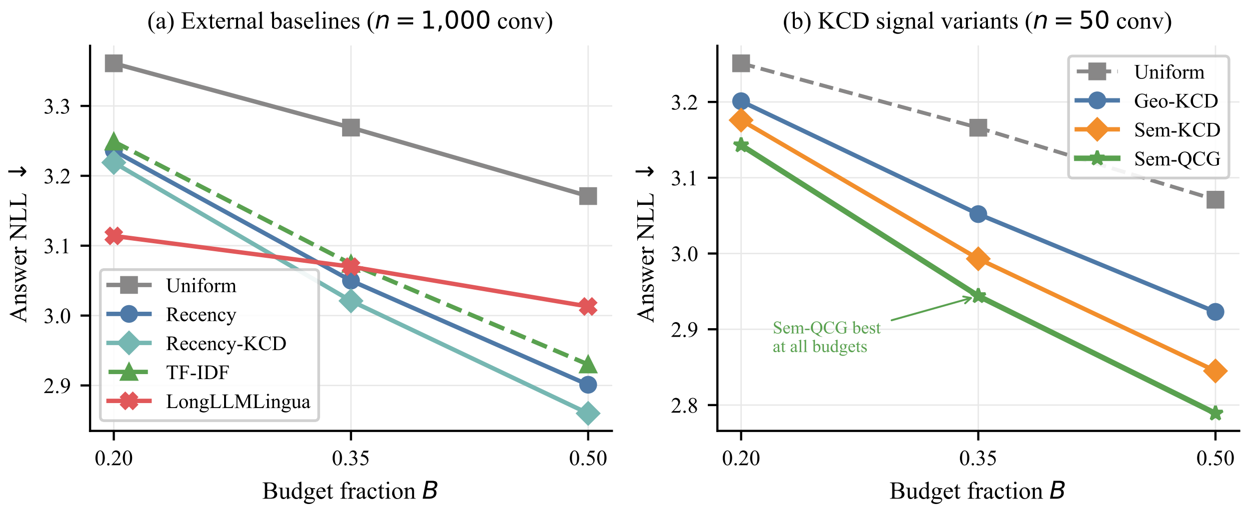

Figure 2: Answer NLL vs. budget. Sem-QCG achieves best behavioral quality at all budgets among KCD variants on MSC (n=1,000 conv), confirming that query-conditioned geometry does not sacrifice behavioral fidelity for geometric gain.

Table 2: kcd signal variant comparison on MSC valid (n=1,000 conversations, Qwen2.5-1.5B). Bold = best per column. Lower is better.

	Logit ℓ_2 ↓			Answer NLL ↓		
Policy	B=0.20	B=0.35	B=0.50	B=0.20	B=0.35	B=0.50
Uniform	799.6	783.0	761.6	3.153	3.061	2.959
geo-kcd	804.9	814.5	812.8	3.105	2.973	2.811
sem-kcd	806.9	814.2	815.0	3.040	2.884	2.716
sem-qcg	807.5	801.8	705.5	3.034	2.850	2.671

4 Signal Variant Comparison

4.1 Setup

We compare the three kcd scoring signals (§2.3) on MSC valid, using 50 conversations and Qwen2.5-1.5B-Instruct. This yields approximately 605 evaluation rows per policy per budget. Statistical significance is assessed by a two-sided signflip permutation test (Ojala and Garriga, 2010), reported at both row level (per evaluation row) and conversation level (per-conversation means aggregated). Conversation-level tests are the primary evidence; row-level tests are reported for completeness but are anti-conservative due to within-conversation dependence.

4.2 Results

Table 2 shows that sem-qcg uniquely resolves the geometry-behavior tradeoff. A key context: all other compression policies are geometrically worse than uniform at every budget—geo-kcd incurs +5.3/+31.6/+51.2 logit ℓ_2 versus uniform at B=0.20/0.35/0.50; sem-kcd incurs +7.3/+31.3/+53.4. Only at budget 0.50 does sem-qcg break this pattern, achieving logit ℓ_2 = 705.5—lower than uniform sampling (761.6)—meaning the compressed context is strictly more faithful to the full-context

Table 3: Pairwise comparison: sem-qcg minus sem-kcd on MSC valid (n=1,000 conversations). Negative Δ means sem-qcg is better. p_{row} = row-level signflip; p_{conv} = conversation-level (primary evidence). CI = 95% bootstrap confidence interval.

Budget	Δ L2 (row)	p_{row}	Δ L2 (conv)	p_{conv}	Δ NLL (conv)	p_{conv}
0.20	+0.61	0.599	−0.15	0.872	−0.011	< 0.001
0.35	−12.47	<0.001	−12.23	<0.001	−0.038	<0.001
0.50	−109.44	<0.001	−105.14	<0.001	−0.044	<0.001

Table 4: Component ablation of sem-qcg on MSC valid (n=1,000 conversations). Removing support-score weighting (w/o support) catastrophically degrades both metrics at all budgets (p<0.001). Removing query projection (w/o query) is marginally better than full sem-qcg at B=0.20 on NLL (p=0.004) but worse at $B \geq 0.35$ on geometry (p<0.001), showing the query projection is beneficial only under moderate or high budget.

	Logit $\ell_2 \downarrow$			Answer NLL $\downarrow$		
Policy	B=0.20	B=0.35	B=0.50	B=0.20	B=0.35	B=0.50
sem-qcg	807.5	801.8	705.5	3.034	2.850	2.671
w/o query proj.	806.7	809.4	708.3	3.028	2.856	2.670
w/o support score	808.9	817.0	757.7	3.094	2.957	2.762

hidden state than randomly retaining half the turns. This is the only regime in our experiments where any compression policy is geometrically more faithful than the no-selection baseline.

Table 3 shows that the improvement is statistically robust at conversation level: at B=0.35, the geometry advantage is Δ=−12.23 logit ℓ_2 ($p_{\text{conv}} < 0.001$) and the behavioral advantage is Δ=−0.038 NLL ($p_{\text{conv}} < 0.001$). At B=0.50 the geometry improvement is dramatic (Δ=−105.1, $p_{\text{conv}} < 0.001$) with 95% CI $[-110.3, -100.2]$.

Mechanism. The query-conditioned projection $\mathbf{u}_t^{(q)}$ biases turn selection toward turns geometrically proximate to the current query's hidden state, independently of surface-level semantic similarity. This is especially useful at higher budgets, where many turns score similarly on semantics but differ in their geometric proximity to the query. The result is that sem-qcg fills budget with turns that are both semantically relevant (from the shortlist) and geometrically load-bearing for the response the model will produce.

Table 4 isolates the two novel components. Removing the support-score weighting (w/o support) is catastrophic: logit ℓ_2 degrades by +52.2 at B=0.50 ($p_{\text{conv}} < 0.001$) and answer NLL worsens by +0.059 ($p_{\text{conv}} < 0.001$), confirming that the support-score term is the load-bearing component of sem-qcg. Removing query-conditioned projection (w/o query) reveals a budget-dependent pattern: at B=0.20, it is marginally better on NLL (Δ=−0.006, p_{conv}=0.004) but at B=0.35 and B=0.50 it is worse on geometry (+7.5 and +3.1 logit ℓ_2, p_{conv}<0.001 and p_{conv}=0.054). The query projection is beneficial when the budget allows meaningful turn differentiation, but contributes noise under very tight budgets where turns compete for few slots.

Table 5: External baseline comparison on the hard stress set (Qwen2.5-1.5B, 9 conversations, 540 rows). Lower is better. *p<0.05, **p<0.01, conversation-level signflip test vs. uniform.

	Logit ℓ_2 ↓			Answer NLL ↓		
Policy	B=0.20	B=0.35	B=0.50	B=0.20	B=0.35	B=0.50
Uniform	387.4	338.0	268.5	1.627	1.272	1.353
Recency	**348.4****	326.3	**274.6**	1.620	1.115	1.027
Recency-kcd	**348.4****	**311.8**	290.9	1.620	1.115	0.959
TF-IDF	381.9	323.1	276.9	1.553	1.004	0.801
LongLLMLingua	385.6	372.4*	369.8**	**1.252****	**0.758***	**0.697****

5 Constraint-Critical Memory

5.1 External Baselines on the Stress Set

We first establish how external compression methods behave on constraint-critical memory, confirming that geometry-behavior divergence is not benchmark-specific.

Table 5 shows that the geometry-behavior divergence documented on MSC (Table 1) replicates on constraint-critical memory. LongLLMLingua achieves the best NLL at every budget—at B=0.50, NLL drops from 1.353 (uniform) to 0.697 (p=0.008, conv)—but degrades logit ℓ_2 by +34.4 at B=0.35 (p=0.017) and by +101.3 at B=0.50 (p=0.004). Recency-based compression achieves the best geometric fidelity at low budget (Δ=−39.0 at B=0.20, p=0.003) with no significant NLL improvement. Geometry-behavior divergence is therefore a property of the compression method, not a property of the benchmark. The signal-swap ablation below shows that geometry-aware turn selection retains the representational structure that constraint-critical answers require, while semantic scoring and LongLLMLingua sacrifice that structure for NLL gain.

5.2 Setup and Motivation

We construct a synthetic stress set of 9 conversations (540 evaluation rows) spanning three families: long-dependency (exact user instructions referenced many turns later), retrieval-heavy (verbatim fact bundles required for correct completion), and code-conversation (code artifacts with precise syntax constraints). These families exercise the hardest case for compression: the signal-to-noise ratio for support turns (turns that contain the constraints) is low, because semantically similar assistant echoes compete with user instructions in any similarity-based ranking.

We test the signal-swap hypothesis: holding the kcd codec fixed, does the choice of scoring signal matter? We compare geo-kcd versus sem-kcd on Qwen2.5-1.5B and Qwen2.5-3B-Instruct.

5.3 Signal-Swap Results

Table 6 confirms that geometry-based scoring is substantially better than semantic scoring on constraint-critical memory. At Qwen2.5-3B, budget 0.50, the NLL damage from using the wrong signal is +1.126 nats (p = 0.0003)—the largest effect size in our entire experimental program. The NLL advantage grows with model scale (1.5B: +0.628 nats; 3B: +1.126 nats), consistent with the interpretation that richer representations make incorrect turn selections more consequential. The logit ℓ_2 advantage is significant at low and mid budgets but attenuates at B=0.50 (where both policies have more budget to include turns and signal quality matters less).

Table 6: Signal-swap ablation on the hard stress set. Δ = sem-kcd minus geo-kcd; positive = geometry is better. p values are conversation-level signflip tests. $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$.

Model	Budget	Δ Logit ℓ_2	p	Δ Answer NLL	p
Qwen2.5-1.5B	0.20	+37.8	0.028*	+0.195	0.001**
	0.35	+38.5	0.079	+0.424	0.011*
	0.50	+0.1	0.995	+0.628	0.001**
Qwen2.5-3B	0.20	+44.9	0.048*	+0.494	0.008**
	0.35	+69.7	0.050*	+0.853	0.005**
	0.50	−29.1	0.128	+1.126	**0.0003*****

Table 7: Support-turn rescue analysis on the hard stress set (36 conversations with identified support turns, budget 0.35). A rescue occurs when a policy retains the latest support turn that uniform sampling would drop.

Outcome	geo-kcd	sem-kcd
Policy rescues more support turns than uniform	17/36	7/36
Uniform beats policy on support turns	2/36	5/36
Constraint turns rescued (from long-dep, code)	13	4
Base-memory turns rescued (persona, fact)	6	3

5.4 Support-Turn Rescue

To understand the mechanism, we trace which turns each policy retains relative to uniform sampling. A support turn is defined as any turn containing a constraint (instruction, code artifact, verbatim fact) that is required for correct completion of the target turn.

Table 7 reveals the mechanism: geo-kcd rescues the latest support turn in 17/36 conversations versus 7/36 for sem-kcd. User instructions produce larger local hidden-state displacement than semantically similar assistant echoes, because the model updates its representational context more substantially when processing novel, directive content than when processing a rephrasing or acknowledgment of prior information. Geometry detects this signature; cosine similarity does not, because the semantic content of an instruction and its echo can be similar while their geometric roles are opposite.

6 Scope Conditions

The results above establish the efficacy of geometry-based scoring under specific conditions. We now characterize two scope conditions that bound these claims.

6.1 Architecture Dependence

On Llama-3.2-3B-Instruct (Table 8), semantic compression significantly outperforms geometry at all budgets for behavioral quality ($p \leq 0.026$ conversation level), and the logit ℓ_2 advantage of semantic is not significant at any budget ($p \geq 0.14$). This is the opposite of the Qwen pattern.

We attribute this to pretraining regime differences. Instruction-following finetuning via RLHF or instruction tuning (Ouyang et al., 2022; Wei et al., 2022) reshapes the model's representational

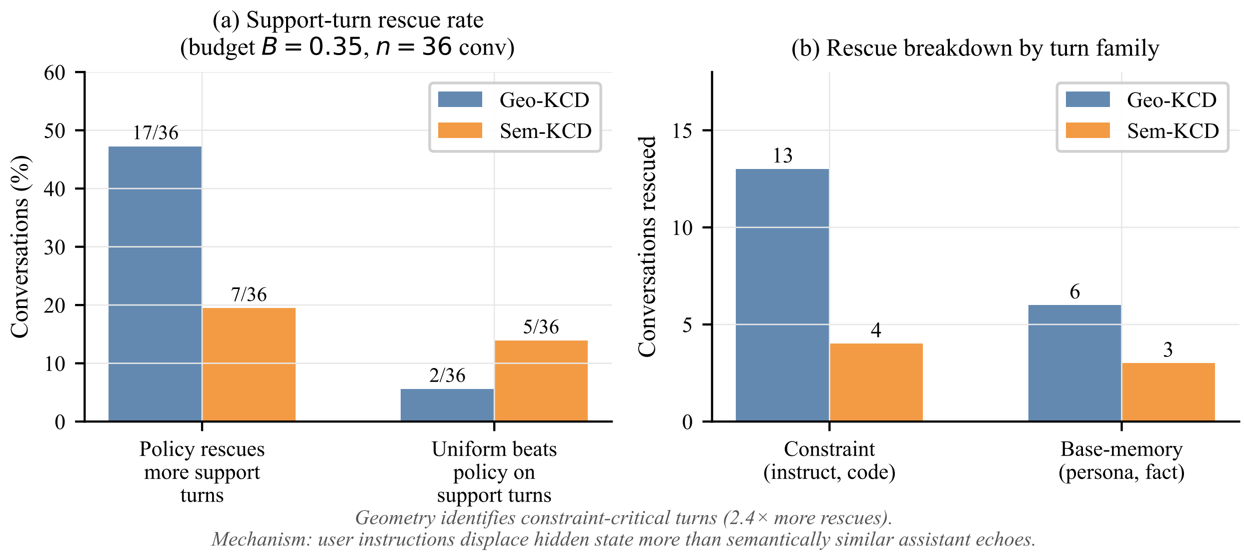

Figure 3: Support-turn rescue analysis on the hard stress set (n=36 conversations, budget B=0.35). Left: overall rescue counts relative to uniform. Right: breakdown by turn family. Geo-KCD rescues 2.4× more constraint-critical turns than Sem-KCD.

Table 8: Cross-architecture comparison on MSC valid (n=50 conversations) with Llama-3.2-3B-Instruct. Lower is better. ΔNLL (conv) = sem-kcd minus geo-kcd at conversation level; negative means sem-kcd is better.

	Logit ℓ_2 ↓			Answer NLL ↓		
Policy	B=0.20	B=0.35	B=0.50	B=0.20	B=0.35	B=0.50
Uniform	689.7	669.4	631.0	3.722	3.657	3.524
geo-kcd	690.6	643.2	592.2	3.617	3.416	3.260
sem-kcd	671.9	633.5	586.1	3.526	3.317	3.139
ΔNLL (conv)	-0.091^{**}	-0.099^{*}	-0.121^{**}	p = 0.007 0.026 0.009		

geometry to respond more sharply to directive content (user instructions, constraints, factual assertions). Qwen2.5 models undergo heavy such finetuning, producing discriminative per-turn geometry where user instructions reliably move the hidden state more than assistant echoes. Llama-3.2-3B's training does not produce this property to the same degree; the geometric signature of constraint-critical turns is weaker, making semantic similarity the more reliable selection criterion. The scope condition is therefore about the model's representational structure induced by its training regime, not parameter count.

6.2 Length Dependence

Table 9 shows results on LongMemEval-S (Wu et al., 2025), a benchmark of episodic retrieval tasks with 400–600 turn conversations (no truncation). All kcd policies improve behavioral quality versus uniform ($p < 0.05$ at conversation level for all NLL comparisons at budgets 0.20–0.35), confirming that turn-level selection generalizes across benchmarks.

However, the geometry advantage does not replicate at conversation level— the pattern fully

Table 9: Cross-benchmark results on LongMemEval-S (n=100 full-length conversations, 400–600 turns, Qwen2.5-1.5B, binding-budget variant: recent-window = 0). Lower is better.

	Logit ℓ_2 ↓			Answer NLL ↓		
Policy	B=0.20	B=0.35	B=0.50	B=0.20	B=0.35	B=0.50
Uniform	1750.7	1781.8	1714.2	0.222	0.191	0.141
geo-kcd	1763.0	1701.3	1683.4	0.135	0.109	0.096
sem-kcd	1750.6	1700.5	1717.9	0.100	0.096	0.094
sem-qcg	1755.9	1748.1	1705.0	0.108	0.105	0.099

reverses. At budget 0.35, sem-qcg is significantly worse than sem-kcd on both metrics: logit ℓ_2 (Δ=+47.6, $p_{\text{conv}} = 0.018$) and answer NLL (Δ=+0.012, $p_{\text{conv}} = 0.001$). At budget 0.50, the L2 difference is not significant ($p_{\text{conv}} = 0.629$), but sem-qcg remains significantly worse on NLL (Δ=+0.006, $p_{\text{conv}} < 0.001$; see Table 11 in the Appendix). sem-kcd achieves the best NLL across all budgets (0.100 at B=0.20, vs. 0.222 for uniform; $p < 0.001$). This reversal has a natural explanation: in 400–600 turn conversations, each turn's geometric signature is competing with hundreds of other turns in a crowded representational space. Per-turn geometric features become noisier as trajectory length grows, because individual displacement vectors represent smaller fractions of the total trajectory variance. Semantic shortlisting, which operates at coarser granularity, remains discriminative; geometry provides no additional value within a crowded shortlist.

Summary. The geometry advantage is reliable for conversations of roughly 20–100 turns (MSC regime): at budget 0.50, sem-qcg beats uniform on both geometry and behavior—the only regime in our experiments where any compression policy is strictly more faithful than no selection. For very long conversations (> 400 turns), the advantage inverts: sem-kcd dominates on both metrics and sem-qcg underperforms it on NLL at all budgets. Signal design should therefore be conditioned on expected conversation length.

7 Related Work

Prompt and context compression. Jiang et al. (2024) propose perplexity-guided token-level compression for long-context LLM inference. We use LongLLMLingua as a primary baseline and show it achieves only marginal NLL improvement while incurring large geometric degradation— evidence that NLL-optimized compression is not sufficient for representational fidelity. Pan et al. (2024) propose LLMLingua-2, a data-distillation approach for task-agnostic prompt compression. Chevalier et al. (2023) use language model fine-tuning to learn compact memory representations; our approach operates at inference time without modifying model weights. He et al. (2025) train a context compressor as a preprocessing step.

A complementary line of work compresses the KV cache at the token level during generation. Zhang et al. (2023) retain “heavy hitter” tokens—those with the largest cumulative attention mass—discarding the rest dynamically. Li et al. (2024) identify which prompt tokens the model will attend to before generation begins and snap the cache accordingly. Cai et al. (2024) vary the cache budget across transformer layers in a pyramidal schedule. Nawrot et al. (2024) learn to merge and evict KV slots during training. All four operate at sub-token granularity within the KV cache; our kcd codec operates at turn granularity and introduces a dual geometric–behavioral evaluation objective that none of these methods consider.

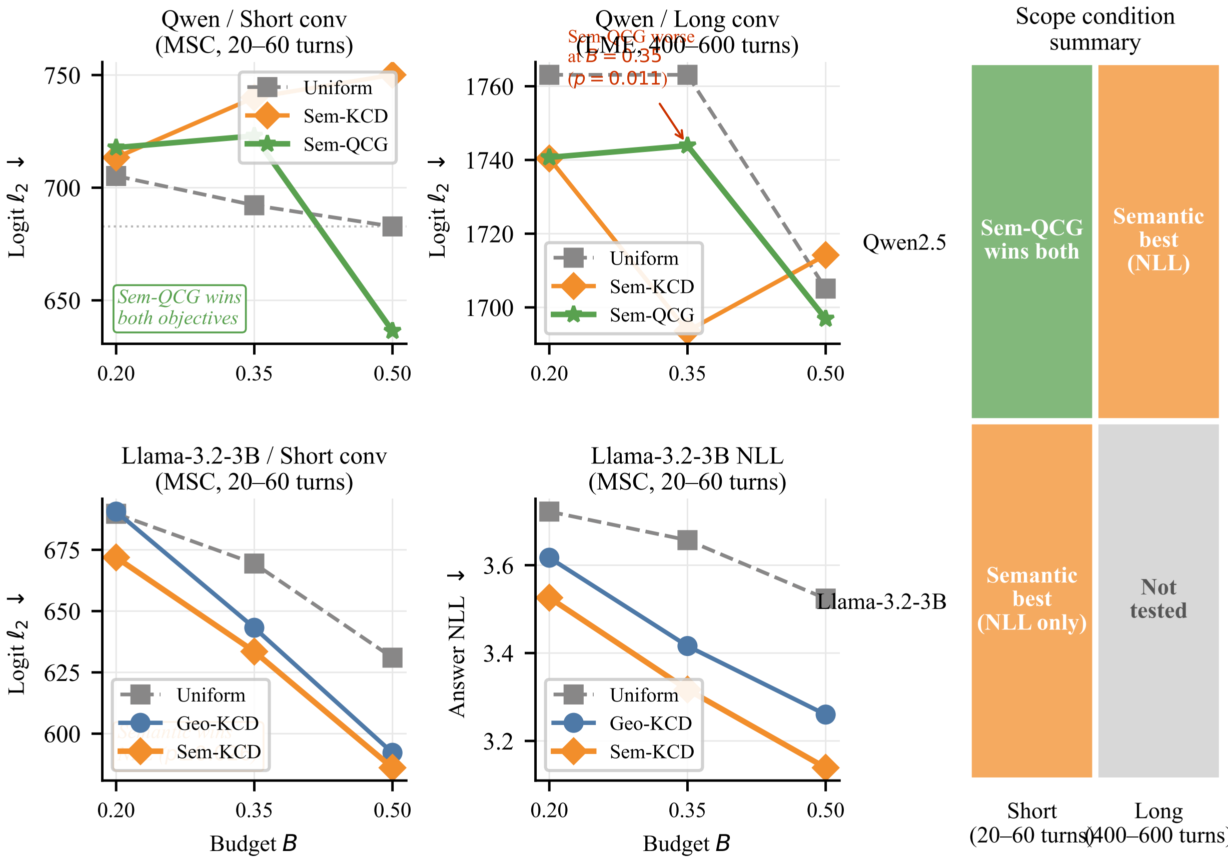

Figure 4: Scope condition map. Rows: model family. Columns: conversation length regime. Green (Sem-QCG wins both): query-conditioned geometry resolves the geometry-behavior tradeoff on Qwen + short conversations. Orange (Semantic best): semantic scoring dominates on Llama and on long conversations where per-turn geometry is noisy. Right panel summarizes the 2×2 grid.

Conversational memory and benchmarks. Xu et al. (2022) introduce Multi-Session Conversations (MSC), a persona-grounded multi-session dialogue dataset that we use as our primary evaluation benchmark. Wu et al. (2025) introduce LongMemEval, testing five core memory abilities across 500 manually curated questions embedded in scalable user-assistant chat histories. Maharana et al. (2024) develop LoCoMo, a benchmark for very-long-term conversational memory. Luo et al. (2026) survey memory mechanisms in LLM agents, classifying them into context-resident compression, retrieval-augmented stores, reflective self-improvement, hierarchical virtual context, and policy-learned management; our kcd codec falls in the context-resident compression family. Nan et al. (2025) propose Nemori, a self-organizing memory system inspired by cognitive science. Wang et al. (2025) introduce HyMem, a hybrid memory architecture with dynamic retrieval scheduling.

Retrieval and external memory. An orthogonal strategy is to externalise history into a retrieval index. Lewis et al. (2020) introduce retrieval-augmented generation (RAG), which retrieves relevant documents at query time rather than compressing the context window. Packer et al. (2023) treat the LLM context window as a fast memory tier and implement page-in/page-out semantics to a slower external store. Our approach is complementary: we compress within the context window at inference time without an external index or modified architecture, and we measure the

Table 10: Geometry characterization across model families. Rank-95: intrinsic dimensionality of conversation trajectories (lower = more compact, normalized by trajectory length). Shuffled Rank-95: control with turn order randomized. ρ: Pearson correlation between per-turn geometric distortion and logit ℓ_2 (H3: geometry predicts decoder drift). All p values from conversation-level permutation tests.

Model	Rank-95	Shuffled	p_{H_1}	ρ	p_{H_3}
Qwen2.5-0.5B	1.049	1.486	< 0.001	0.989	< 0.001
Qwen2.5-1.5B	1.167	1.458	0.001	0.994	< 0.001
SmolLM2-1.7B	1.528	1.799	0.012	0.989	< 0.001

geometric cost of compression decisions—a cost that retrieval-based systems incur implicitly when summarising or dropping history.

Hidden-state geometry in language models. Transformer representations are known to occupy a compact, anisotropic subspace. Ethayarajh (2019) show that contextualized embeddings are far from isotropic: upper-layer representations are highly context-specific, with less than 5% of variance explained by static word identity. Cai et al. (2021) further characterize the cluster structure of this embedding space and introduce a rigorous isotropy measure. These findings motivate our Rank-95 metric, which quantifies how compactly conversation trajectories occupy $\mathbb{S}^{d-1}$; the theoretical basis for such dimensionality measures is established in Li et al. (2018) and Aghajanyan et al. (2021). Huh et al. (2024) observe cross-model convergence of representations toward a shared geometric structure, motivating our cross-architecture comparisons. Xiao et al. (2024) identify attention sinks—tokens that accumulate disproportionate attention mass—which we correct for in hidden-state extraction. Anagnostidis et al. (2022) provide theoretical grounding for the low Rank-95 values we observe (1.05–1.53). To our knowledge, we are the first to use parallel-transported geometry on conversation-state trajectories as a turn-selection signal.

8 Geometry Characterization

We present supporting evidence for our geometric motivation.

Table 10 shows that: (H1) Real conversation trajectories are significantly more compact than shuffled controls ($p \leq 0.012$), confirming temporal ordering creates compact geometry. (H3) Per-turn geometric distortion tightly predicts decoder drift ($\rho = 0.989$–0.994, $p < 0.001$), confirming geometric distortion is a meaningful proxy for representational harm. These properties hold across architectures and parameter scales.

9 Limitations

Architecture dependence. The geometry advantage is not universal. On Llama-3.2-3B-Instruct, semantic compression significantly outperforms geometry-based compression for behavioral quality ($p \leq 0.026$, all budgets). We attribute this to pretraining differences, but do not test this hypothesis directly. Practitioners should validate geometric scoring on their target model family before deployment, using a held-out development set.

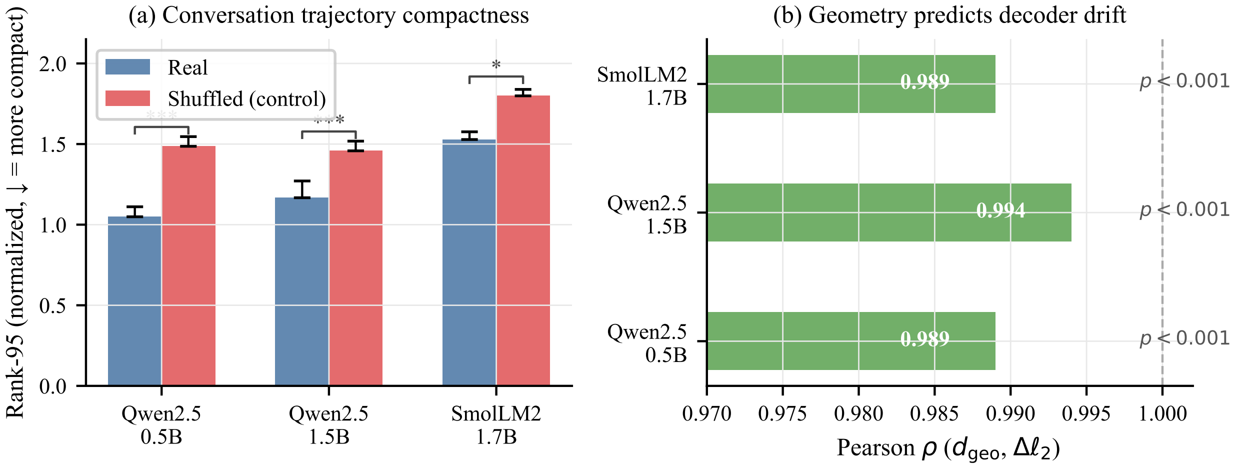

Figure 5: Geometry characterization across model families. Left: Rank-95 for real vs. shuffled conversation trajectories. Real trajectories are significantly more compact ($p \le 0.012$). Right: Pearson ρ between per-turn geometric distortion and logit ℓ_2 ($\rho = 0.989$–0.994, all $p<0.001$).

Conversation length dependence. The geometry advantage degrades on 400–600 turn conversations (LongMemEval-S benchmark): sem-qcg is significantly worse than sem-kcd on logit ℓ_2 at B=0.35 ($p_{\text{conv}} = 0.011$). Per-turn geometric features become less discriminative as trajectory length grows. Our MSC experiments use conversations of 20–60 turns; results may not generalize to much longer sessions without adaptation.

Oracle metric circularity. Our support-turn rescue analysis identifies support turns by manual annotation on a small synthetic stress set (36 conversations). This is a qualitative mechanism study, not a large-scale validation. The claim that geometry detects constraint-critical turns rests on this small-n analysis plus the signal-swap effect sizes in Table 6.

QA evaluation on LME. We ran a QA proxy evaluation on LME using the next conversation turn as the gold answer. Token F1 scores were uniformly near-zero (≈ 0.08) across all policies, and we determined this metric is invalid because LME's conversational gold turns are not faithful proxies for the question-answer pairs in the official evaluation protocol. We therefore do not report these results as primary evidence and rely on NLL and logit ℓ_2 for the LME cross-benchmark analysis.

Scale. Our experiments use models up to 3B parameters. We do not test 7B+ models due to compute constraints. The observed scale trend (NLL damage grows 1.5B $\rightarrow$ 3B) suggests geometry may matter more at larger scale, but this hypothesis requires direct validation.

10 Conclusion

We introduced the kcd codec for conversational memory compression and showed that the choice of turn scoring signal is a first-class design decision. Geometry and behavioral quality are distinct objectives that diverge in practice: LongLLMLingua, a state-of-the-art prompt compression method, trades geometric fidelity for marginal NLL improvement. Query-conditioned geometry inside a semantic shortlist (sem-qcg) resolves this tradeoff on Multi-Session Conversations: it achieves lower logit ℓ_2 than uniform random sampling while simultaneously improving answer NLL, the only policy in our study to win on both objectives simultaneously. On constraint-critical memory, geometry-based scoring substantially outperforms semantic scoring, with the NLL damage from the wrong signal growing from 0.63 to 1.13 nats as model scale increases from 1.5B to 3B. We characterize architecture and length as scope conditions, and provide mechanism evidence through support-turn rescue analysis. We hope these findings motivate dual-objective evaluation and signal-conditioned design as standard practices in conversational memory research.

Acknowledgments

Experiments were conducted on cloud GPU infrastructure. Code, data, and experiment logs are available at https://github.com/SteveMama/rt-geometry-memory.

The authors used Claude (Anthropic) to assist with manuscript drafting and editing. All experimental results, scientific claims, and final text were verified and approved by the authors.

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A Experimental Details

A.1 Models and Hardware

All experiments run on Qwen2.5-1.5B-Instruct, Qwen2.5-3B-Instruct (Qwen Team, 2025), and Llama-3.2-3B-Instruct (Grattafiori et al., 2024). Models are loaded in bfloat16 with Flash Attention 2 (Dao et al., 2022) on NVIDIA L40S GPUs (46 GB VRAM). All MSC experiments run with 7 GPUs and 2 workers per GPU to saturate VRAM (≈ 22 GB per worker).

A.2 Dataset Details

MSC valid: Multi-Session Conversations validation split (Xu et al., 2022). Conversations average 5 sessions of 4–6 turns each. All 1,000 conversations used for baseline comparison; 1,000 conversations for signal variant study (full validation split).

LongMemEval-S: Wu et al. (2025) benchmark. 100 full-length conversations, 400–600 turns each. No truncation applied (evaluating compression policies, not context window limits).

Hard stress set: 9 synthetic conversations across three families (long-dependency, retrieval-heavy, code-conversation), 540 evaluation rows, hand-constructed by the authors to maximize signal-to-noise difficulty for support-turn detection.

A.3 Compression Parameters

Budget fractions: $B \in \{0.20, 0.35, 0.50\}$. KCD compression ratio: $\alpha = 0.5$ (compressed turns retain half their tokens, selected by leading-token truncation). Semantic shortlist size for sem-qcg: top-k turns where $k = \lceil 2B \cdot T \rceil$ (twice the keep budget, so geometry selects the final $B \cdot T$ from $2B \cdot T$ candidates).

A.4 Statistical Protocol

All significance tests are two-sided signflip permutation tests with 5,000 resampling iterations. We report both row-level (anti-conservative) and conversation-level (conservative, primary) p-values. No multiple-comparison correction is applied within individual tables; results are interpreted directionally, and effect sizes are reported alongside p-values.

A.5 Geometry Feature Extraction

Hidden states are extracted from the final transformer layer, normalized to the unit sphere. Parallel transport uses the exponential and logarithm maps on $\mathbb{S}^{d-1}$:

$$\mathrm{Log}_{\mathbf{p}}(\mathbf{q}) = \frac{\arccos(\mathbf{p}^\top \mathbf{q})}{\sqrt{1 - (\mathbf{p}^\top \mathbf{q})^2}} (\mathbf{q} - (\mathbf{p}^\top \mathbf{q})\mathbf{p})$$

Reference point $\mathbf{h}_{\text{ref}}$ is the Fréchet mean of the trajectory, approximated by iterative geodesic midpoints. Attention weights (for the attention-augmented feature set) are averaged over the final 4 transformer layers with sink correction: the first 32 tokens are excluded from weight normalization following Xiao et al. (2024).

B Additional Pairwise Results

B.1 LME Pairwise: sem-qcg vs. sem-kcd

B.2 Llama-3.2-3B Pairwise: sem-kcd vs. geo-kcd

Table 11: sem-qcg minus sem-kcd on LongMemEval-S (n=100 conversations). Positive Δ logit L2 means sem-qcg is geometrically worse.

Budget	Δ L2 (row)	p_{row}	Δ L2 (conv)	p_{conv}	Δ NLL (conv)	p_{conv}
0.20	+5.2	0.476	+5.2	0.742	+0.010	0.738
0.35	+47.6	<0.001	+47.6	0.018	+0.012	0.001
0.50	−12.9	0.108	−12.9	0.629	+0.006	<0.001

Table 12: sem-kcd minus geo-kcd on Llama-3.2-3B (n=50 conversations). Negative Δ NLL means sem-kcd is better behaviorally.

Budget	Δ L2 (conv)	p_{conv}	Δ NLL (conv)	p_{conv}
0.20	−18.7	0.141	−0.091	**0.007**
0.35	−9.7	0.514	−0.099	**0.026**
0.50	−6.1	0.778	−0.121	**0.009**